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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Nabil The Goat
Selected Domain	Domain 2: SME automation for Orbyte (sponsored by Alpha Red Solutions)
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth (secondary: SDG 9: Industry, Innovation and Infrastructure)
Project / Solution Name	sabave.ai
Name (Leader)	Nabil Hilmi Tsaqifi
Email & Phone (Leader)	nabiltsaqifi@gmail.com / +60 11 7244 6398
University Affiliation(s)	Monash University Malaysia
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's micro, small and medium enterprises (MSMEs) generate 39.5% of GDP and 48.7% of employment (DOSM, 2024), yet most run their marketing by hand through WhatsApp, Facebook and spreadsheets. When bad weather hits, especially transboundary haze and heavy rain or thunderstorms, walk-in customers stay home and revenue falls. The October 2026 haze is estimated to cost Malaysia RM20–50 million a day, with slower SME retail activity among the effects (The Star, 14 Oct 2026), and businesses that rely on consumer foot traffic are hit hardest (The Star, 21 Oct 2026).

In our conversations with local SME owners during the current haze, owners reported falling revenue but said they mostly accept it, and some consider laying off staff to cut costs. They do not respond with marketing because spotting the forecast, choosing an offer, designing a poster and writing bilingual captions takes hours they do not have, so any reaction comes after the damage is done. Larger competitors with marketing teams react days earlier. This information and capacity gap widens the divide between small and large businesses.

The problem is recurring: haze and monsoon rains return every year, and the burden is the same for cafés, laundromats, retailers and other walk-in businesses. Lost weeks of sales squeeze thin margins, and cost-cutting through layoffs directly threatens decent work (SDG 8).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

sabave.ai (save + brave) is a weather-triggered marketing automation module for Orbyte, the shared business hub for Malaysian SMEs. It turns a weather forecast into a ready-to-approve, on-brand campaign before the bad weather arrives.

1. Trigger: sabave.ai checks each business's location against live Open-Meteo forecasts for haze (PM2.5) and heavy rain or thunderstorms. Crossing a threshold creates an alert. **2. Connected data:** it loads the business's Brand Kit from the hub: logo, colours, past poster style, website voice, ethos, products and prices, target customers, channels, language, a “never say” list and a budget cap. **3. Decision logic:**

transparent rules select 2–3 campaign options (e.g. delivery push, stay-home bundle, rain-check voucher), each with a suggested budget and a plain-language reason. Generative AI (a DeepSeek LLM) is used only where it adds value: writing Bahasa Malaysia and English captions in the brand's voice and summarising brand style. Every output is checked against the never-say list and a health-claim filter, with a template fallback. **4. Human approval:** staff edit the draft and poster; only the Owner role can approve, within the budget cap, enforced on the server. **5. Visible outcome:** the campaign is published to Facebook/Instagram through a connector (simulated in the prototype), and every step is recorded in an append-only activity log.

Weather signals are pluggable, so new triggers such as heatwaves can be added without changing the journey, and the social connector is built as a reusable Orbyte service that any module can call.

From the SDC Phase 1 workshops we apply GenAI APIs (server-side LLM calls with structured JSON output), database architecture (business, campaign and audit-log data models designed for a multi-tenant hub, with Firestore as the production database) and cloud hosting (deployment and secrets management on Replit).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS. **Backend:** Next.js API routes on Node.js 22. **Data:** synthetic JSON seed data and an in-memory store for the prototype; Google Firestore planned for production. **APIs:** Open-Meteo Air Quality API (PM2.5) and Open-Meteo Weather Forecast API (rain, thunderstorms); DeepSeek API (latest Pro model, OpenAI-compatible) for copywriting and brand-style summaries; a mock Meta (Facebook/Instagram) connector behind a swappable interface. **Hosting:** Replit (development, Secrets, deployment); GitHub for source control.

System block diagram:

User (browser: Staff / Owner) → Next.js UI on Replit → Next.js API routes → [Weather Signal Engine → Open-Meteo APIs] → [Trigger Engine → Campaign Rules] → [Copy Generator → DeepSeek API, with template fallback] → [Approval Service: role, budget and never-say checks] → [Social Connector → Meta (mocked)] → [Activity Log] → result shown to the user.

Sponsor resources: Yes. Replit (Replit Core) is used to develop, store secrets for and deploy the live prototype. The RunPod credit was not used because the project has no GPU workload; AI inference is through the DeepSeek API.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we spoke with local SME owners during the October 2026 haze. They described revenue drops and a sense that nothing could be done. **Define:** the real burden is not only missing information but the hours needed to turn a forecast into a campaign. **Ideate:** we first considered a prediction dashboard but rejected it because it adds reading work without removing any task, so we redesigned it so that the forecast triggers ready-made, on-brand campaigns. **Prototype:** a working Next.js app with seeded synthetic businesses, simulate buttons for demos, and fallbacks for every external dependency so the journey never breaks. **Test:** each build stage is tested end to end before it is committed.

Deployment readiness: the prototype is live on Replit with keys in Replit Secrets. For Phase 2 we will test with the SME owners we interviewed, measure time-to-campaign, refine rules and copy from their

feedback, and add Firestore persistence and scheduled forecast checks. **Biggest risk:** Meta API app review can take weeks. **Fallback:** the mock connector, plus a downloadable poster and caption that owners can post manually.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.3**: promote development-oriented policies that support productive activities, decent job creation and entrepreneurship, and encourage the growth of micro-, small- and medium-sized enterprises. **Target 8.2**: achieve higher levels of economic productivity through diversification, technological upgrading and innovation. Secondary: **Target 9.3**, integration of small-scale enterprises into value chains and markets.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• Most SME owners we spoke to took **no marketing action** during the haze (team interviews, Oct 2026). • An estimated **~90 minutes** of owner time to produce one weather-reactive campaign manually: checking the forecast, choosing an offer, designing a poster, writing bilingual captions and posting (team estimate). • Haze is estimated to cost Malaysia **RM20–50 million per day** (Yeah Kim Leng, cited in The Star, 14 Oct 2026).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• Minutes from weather alert to approved campaign. • Days of lead time between the approved campaign and the weather event. • % of weather alerts converted into an approved campaign.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• **≥85% reduction** in time per campaign (from ~90 to ≤10 minutes). • Campaigns approved **≥2 days** before the weather event. • Tested with **≥3 real SME owners**, with **≥80%** of alerts converted into an approved campaign.

Sources:

DOSM (2025), MSMEs Performance 2024, reported by Human Resources Online: https://www.humanresourcesonline.net/msmes-in-malaysia-employed-nearly-half-the-workforce-in-2024-with-productivity-on-the-rise-dosm

The Star (14 Oct 2026), "Billion-ringgit cost of the haze": https://www.thestar.com.my/news/nation/2026/10/14/billion-ringgit-cost-of-the-haze

The Star (21 Oct 2026), "Haze clouds outlook": https://www.thestar.com.my/business/business-news/2026/10/21/haze-clouds-outlook

United Nations SDG targets: https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: N/A.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 14	Code (framework)	MIT
React 18	Code (library)	MIT
TypeScript	Code (language)	Apache 2.0
Tailwind CSS	Code (library)	MIT
Node.js 22	Code (runtime)	MIT
Open-Meteo forecast data	Dataset (API)	CC BY 4.0
DeepSeek latest Pro model (via API)	Model (pre-trained)	DeepSeek API Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25 October 2026